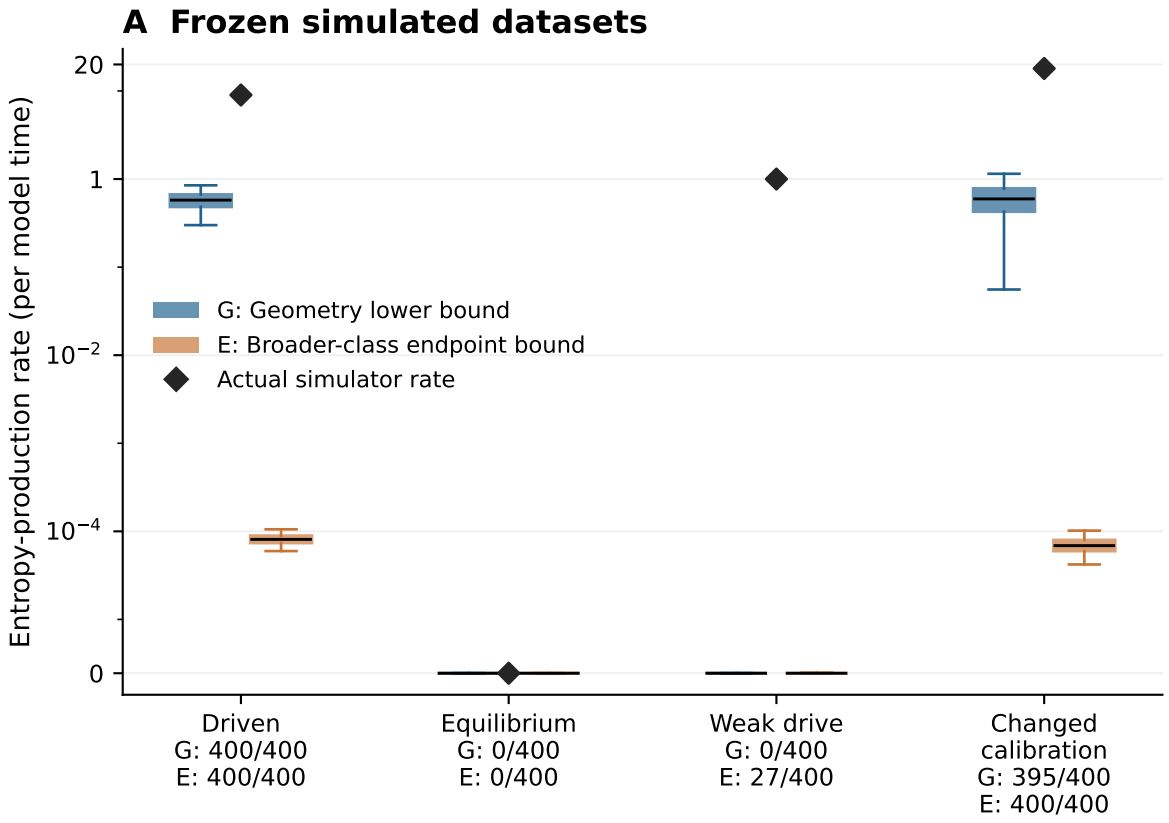
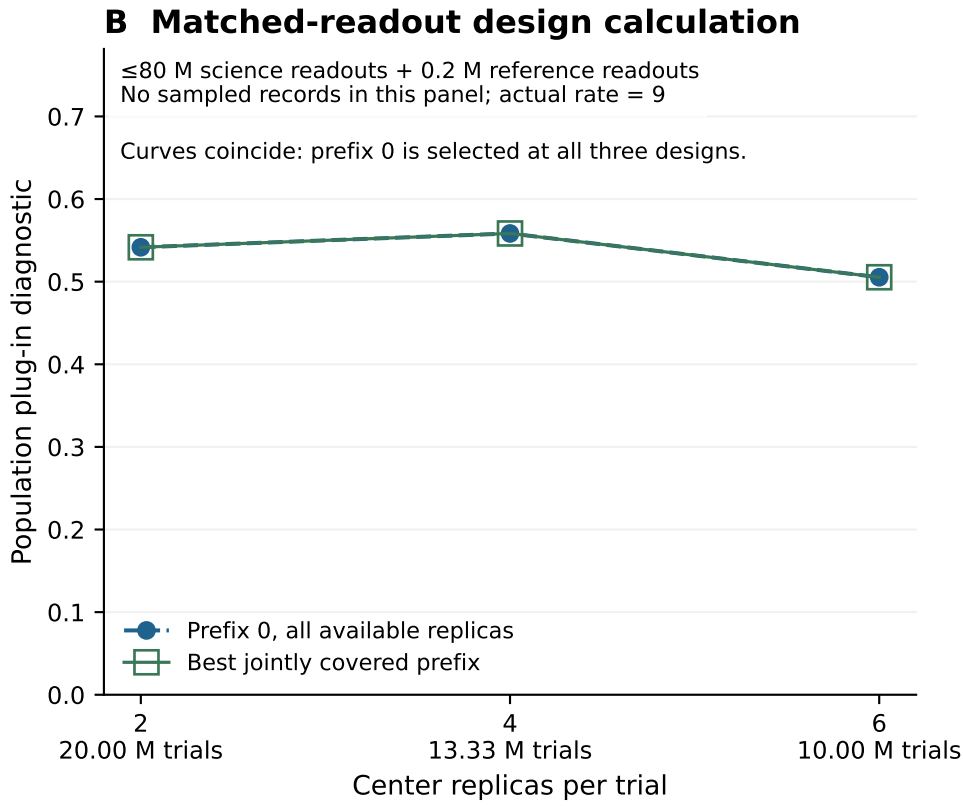
Detector calibration: finite-data behavior and design limits





A: Boxes show 25–75% of 400 simulated datasets per case; whiskers show 2.5–97.5%. Positive counts are shown for each bound.
Geometry uses a pointwise diffusion identity; the endpoint comparison applies to a broader model class. The vertical scale is nonlinear.

B: Expected sufficient statistics are inserted into the confidence formulas. These values are neither confidence certificates nor power estimates.
The pilot splits failure probability 0.025/0.025; the frozen benchmark uses 0.04/0.01. Preparation, calibration infrastructure and energy are additional costs.